CLASS-8

LESSON-5 SQUARES AND SQUARE ROOTS

(This PDF Based on NCERT Book)

PROPERTIES OF SQUARE NUMBERS(वर्गसंख्या के गुण)-

When you look at the last digit of a square number, there are very specific patterns. Here is how it works for numbers ending in **1** and **6**:

1. Square Numbers ending with 1

A square number ends in **1** only if the original number being squared ends in either **1** or **9**.

- **If the number ends in 1:**
  - $1 \times 1 = 1$
  - $11 \times 11 = 121$
  - $21 \times 21 = 441$
- **If the number ends in 9:**
  - $9 \times 9 = 81$
  - $19 \times 19 = 361$
  - $29 \times 29 = 841$

2. Square Numbers ending with 6

A square number ends in **6** only if the original number being squared ends in either **4** or **6**.

- **If the number ends in 4:**
  - $4 \times 4 = 16$
  - $14 \times 14 = 196$
  - $24 \times 24 = 576$
- **If the number ends in 6:**
  - $6 \times 6 = 36$
  - $16 \times 16 = 256$
  - $26 \times 26 = 676$

Last digit of Number	Last digit of its Square
1 or 9	**1**
2 or 8	**4**
3 or 7	**9**

Last digit of Number	Last digit of its Square
4 or 6	**6**
5	**5**
0	**0**

3. Square Numbers ending with 0

A square number ends in **0** only if the original number being squared ends in either **0**.

When you square a number that ends in zeros, the resulting square number will have **twice** as many zeros at the end.

Ex- $10^2 = 10 \times 10 = 100$

$400^2 = 400 \times 400 = 160000$

<u>SOME MORE INTERESTING PATTERNS (कुछ और दिलचस्प पैटर्न)-</u>

1. Adding Triangular Numbers(त्रिभुजाकर संख्याओ का योग)

Triangular numbers are numbers that can form an equilateral triangle (1, 3, 6, 10, 15...). If you add two **consecutive** triangular numbers, you always get a square number!

- $1 + 3 = 4$ (2^2)
- $3 + 6 = 9$ (3^2)
- $6 + 10 = 16$ (4^2)

2. Numbers Between Two Square Numbers(दो वर्ग संख्यों के बीच की संख्या)

There is a simple formula to find how many **non-square** numbers sit between two perfect squares. Between n^2 and $(n+1)^2$, there are exactly 2n numbers.

- **Example:** Between 3^2 (9) and 4^2(16):
  - $n = 3$, so $2 \times 3 = 6$.
  - The non-square numbers are: 10, 11, 12, 13, 14, 15. (Exactly **6** numbers).

3. Adding Consecutive Odd Numbers(लगातार विषम संख्याओ का योग)

A square number is always the sum of the first n odd numbers starting from 1.

- $1^2 = 1$

- $2^2 = 1 + 3 = 4$
- $3^2 = 1 + 3 + 5 = 9$
- $4^2 = 1 + 3 + 5 + 7 = 16$

4. A Sum of Consecutive Natural Numbers(क्रमानुसार प्राकृतिक संख्याओं का योग)

Some square numbers can be expressed as the sum of consecutive integers. This is most common with odd square numbers. Any odd square number n^2 can be expressed as the sum of two consecutive natural numbers:

$$\frac{n^2-1}{2}+\frac{n^2+1}{2}$$

Example: $3^2 = 9$. The two numbers are $\frac{9-1}{2}$ =4 and $\frac{9+1}{2}$ =5.

  - So, 9 = 4 + 5.
- **Example:** $5^2 = 25$. The numbers are 12 and 13.
  - So, 25 = 12 + 13.

5. Product of Two Consecutive Even or Odd Numbers(दो लगातार सम या विषम संख्याओ का योग)

If you multiply two consecutive even numbers (like 4 and 6) or two consecutive odd numbers (like 5 and 7), the result is always **one less than** the square of the number between them.

- **Consecutive Even:** 4 × 6 = 24. (The number between them is 5, and $5^2 = 25$. Notice 25 - 1 = 24).
- **Consecutive Odd:** 11 × 13 = 143. (The number between them is 12, and $12^2 = 144$. Notice 144 - 1 = 143).
- **Formula:** $(n-1)(n+1) = n^2 - 1$

6. Palindrome Squares (The 111 Pattern)

When you square numbers made only of the digit **1**, the result is a "palindrome" (it reads the same forward and backward). It creates a perfect mountain of numbers:

- $1^2 = 1$
- $11^2 = 121$
- $111^2 = 12321$
- $1111^2 = 1234321$

FINDING THE SQUARE OF A NUMBER(किसी संख्या का वर्ग ज्ञात करना)-

1. Using the $(a + b)^2$ Formula

If you have a two-digit number, you can split it into tens and units. The formula is:

$a^2 + 2ab + b^2$

Example: 23^2 (Think of this as 20 + 3)

- a = 20, b = 3
- $a^2 = 400$
- 2ab = 2 × 20 × 3 = 120
- $b^2 = 9$
- **Add them up:** 400 + 120 + 9 = {529}.

2. **PYTHAGOREAN TRIPLETS**

Method 1: The "Odd Number" Starting Rule

If you want to find a triplet starting with an **odd number** (n), use this simple trick:

1. **Pick an odd number** (n). Let's use **3**.
2. **Square it:** $3^2 = 9$.
3. **Split that result into two consecutive numbers** (two numbers next to each other that add up to 9): **4 and 5**.
4. **Your Triplet:** (3, 4, 5).

Another Example with 7:

1. **Pick 7.**
2. **Square it:** $7^2 = 49$.
3. **Split 49 into two consecutive numbers:** 24 + 25 = 49.
4. **Your Triplet:** (7, 24, 25).

Method 2: The General Formula (For any number)

If you have any number (m) greater than 1, you can use these three formulas to get the three sides of the triplet:

- **Side 1:** 2m
- **Side 2:** $m^2 - 1$
- **Side 3:** $m^2 + 1$ (This will be the largest side)

Example: Let's pick m = 4

- **Side 1:** 2 × 4 = {8}
- **Side 2:** $4^2 - 1 = 16 - 1$ = {15}
- **Side 3:** $4^2 + 1 = 16 + 1$ = {17}
- **Your Triplet:** (8, 15, 17).

Method 3: The "Scaling" Method

Once you know one triplet, you can find infinite others just by **multiplying** all three numbers by the same amount.

Take the basic **(3, 4, 5)** triplet:

- Multiply by 2: **(6, 8, 10)** → (36 + 64 = 100)
- Multiply by 10: **(30, 40, 50)** → (900 + 1600 = 2500)

If the number is...	Use this trick	Example
Odd	Square it, then split into neighbors	5 $\rightarrow$ 25 $\rightarrow$ **(5, 12, 13)**
Even	Use the 2m, m^2-1, m^2+1 formula	m=3 $\rightarrow$ **(6, 8, 10)**
Known	Just double or triple a triplet you know	(3,4,5) × 3 = **(9, 12, 15)**

FINDING SQUARE ROOTS(वर्गमूल ज्ञात करना)-

. The Repeated Subtraction Method (For Perfect Squares)

This is a fun trick based on the property that every square number is the sum of consecutive odd numbers. To find the root, you keep subtracting odd numbers (1, 3, 5, 7, ...) until you reach **0**. The number of times you subtract is your answer.

Example: Find $\sqrt{16}$

1. 16 - 1 = 15 (Step 1)
2. 15 - 3 = 12 (Step 2)
3. 12 - 5 = 7 (Step 3)
4. 7 - 7 = 0 (Step 4)

- You subtracted **4 times**, so $\sqrt{16}$ = 4.

2. Prime Factorization Method

This is best for larger numbers that you can't solve in your head. You break the number down into its smallest "building blocks" (prime numbers).

Example: Find $\sqrt{144}$

1. Break it down: 144 = 2 × 2 × 2 × 2 × 3 × 3.
2. **Make pairs** of the same numbers: (2 × 2), (2 × 2), (3 × 3).
3. Take **one number from each pair** and multiply them: 2 × 2 × 3 = 12.

- So, $\sqrt{144}$ = 12.

3. The Estimation Method (For Non-Perfect Squares)

If a number isn't a perfect square (like 10), you have to guess and check by finding the "neighbors."

Example: Find $\sqrt{10}$

1. Find the nearest squares: $3^2 = 9$ and $4^2 = 16$.
2. Since 10 is between 9 and 16, the answer must be between 3 and 4.
3. Since 10 is very close to 9, the answer is likely something like **3.1 or 3.2**.

4. The Long Division Method

This is the "pro" method used for finding the exact square root of any number, even decimals. It looks a bit like long division but with a twist.

1. **Group the digits** in pairs from right to left (e.g., 5,29 becomes 5 and 29).
2. Find the largest square less than the first group ($2^2 = 4$).
3. Subtract and bring down the next pair.
4. Double the quotient and find a new digit to continue.

Shutterstock
Explore

Summary Table

Method	**Best Used For...**	**Difficulty**
Repeated Subtraction	Small perfect squares (like 25, 36)	Easy
Prime Factorization	Medium numbers (like 144, 400)	Medium
Estimation	Numbers that aren't perfect (like 20)	Easy (Approximate)
Long Division	Any number / Decimals	Hard